

How to express  $P_k + jQ_k$  as a fct of  $\bar{V}$ ?

$$\underline{P} + j\underline{Q} = \underline{\bar{V}} \circ \underline{\bar{I}}^* = \underline{\bar{V}} \circ \underline{Y}^* \underline{\bar{V}}^*$$

Let's expand this for mode  $k$ :

$$P_k + jQ_k = \bar{V}_k \times (\underline{Y}^* \underline{\bar{V}}^*)_k$$

$$= \bar{V}_k \left( y_{kk}^* \bar{V}_k^* + \sum_{m \in N_k} y_{km}^* \bar{V}_m^* \right)$$

$$= y_{kk}^* V_k^2 + \bar{V}_k \sum_{m \in N_k} y_{km}^* \bar{V}_m^*$$

$$= (g_{kk} - jb_{kk}) V_k^2 + \sum_{m \in N_k} (g_{km} - jb_{km}) V_k V_m \underbrace{e^{j(\theta_k - \theta_m)}}_{\cos(\theta_k - \theta_m) + j \sin(\theta_k - \theta_m)}$$

$$\cos(\theta_k - \theta_m) + j \sin(\theta_k - \theta_m)$$

$$\Rightarrow P_k = g_{kk} V_k^2 + \sum_{m \in N_k} V_k V_m (g_{km} \cos(\theta_k - \theta_m) + b_{km} \sin(\theta_k - \theta_m)),$$

$$Q_k = -b_{kk} V_k^2 + V_k \sum_{m \in N_k} V_m (g_{km} \sin(\dots) - b_{km} \cos(\dots)).$$